

IDSS Revision Lab

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Note: *These practice questions have been designed to help you prepare for the IDSS exam. Use them to ask questions during the revision lab and to discuss solutions with your classmates. The actual exam questions will be different in content, style, and length, and they may cover a different part of the taught material altogether. Please use these exercises to understand the material and practice the concepts and methods. Do not memorise them or their solutions.*

Linear Algebra

1. Consider vectors $v_1 = (2, 5)$, $v_2 = (-1, 3)$ and $v_3 = (-1, -8)$.
 - (a) Calculate their covariance matrix.
 - (b) Compute the corresponding whitening matrix W using eigen-decomposition.

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Optimisation

2. **Hint:** For the following questions, you can use $((10)(01))^{-1}$ to represent $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1}$ to simplify your typing.

You are given the following linear least squares optimisation problem:

Minimise the cost function:

$$f(x) = \|Ax - b\|^2$$

where:

- $x \in \mathbb{R}^2$ is the vector of unknowns,
- $A \in \mathbb{R}^{2 \times 2}$ is a matrix of known values,
- $b \in \mathbb{R}^2$ is a vector of known values.

Given:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

- (a) Solve the least squares problem using the *normal equations method* to find the optimal solution x^* .

Hint: The normal equations are derived from the gradient of the least squares function, set to zero: $A^\top Ax = A^\top b$. And $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

- (b) Solve the least squares problem using *gradient descent*, starting from an initial guess $x_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and using a step size $\delta = 0.5$. Perform *two iterations*.

Hint: The gradient of the least squares cost function is given by $\nabla f(x) = 2A^\top(Ax - b)$. You can use ∇ to represent ∇ .

Probability Theory

3. This question is concerned with the card game of blackjack, the rules of which are summarised below.

- The game is played with a pack of 52 cards, containing four copies of the values 1 to 10, Jack (J), Queen (Q), King (K).
 - The cards numbered 1 to 10 are worth as many points as their value (so 1 to 10 points respectively).
 - The cards J, Q and K are worth 10 points each
 - Each player receives two cards, and can ask for more.
 - The aim of the game is to get cards with a total value as close to 21 as possible, but no higher.
- (a) If you already have a King and a Queen, only a 1 would bring you to 21 points. What is the probability of this happening?
- (b) You have a 7 and an 8. If you draw another card, what is the probability for you to get a total over 21 and loose?
- (c) What is the expected value of a card drawn from the complete pack of cards?
- (d) You observe another player, and you can see that one of their card is a King, but the second card is hidden. They decide to pass (not draw a card). In previous games, you have observed that:
- 1) they passed 90% of the time when they had 20 points
 - 2) they passed only 20% of the time when they had less than 20 points
- what is the probability that they have 20 points in total?

Databases & Indexing

4. Consider the following relation Book(bookID, title, authorId):

Field	Size
bookId	8 bytes
title	48 bytes
authorId	8 bytes

Also consider the following properties of the Book relation:

- Record size: 64 bytes
- Number of records: 120,000 records
- Block size: 4096 bytes
- Pointer size: 8 bytes

(a) Compute the blocking factor and the number of blocks for Book

(b) Design a primary index on BookID. (This includes the number of index entries, index blocking factor, and number of index blocks.)

(c) Compute the number of levels in a multilevel index, if it's implemented as a sparse index.

(d) Calculate the expected I/O cost for searching for a record using (i) Linear search, (ii) Binary search and (iii) the multi-level index.

(e) Suppose there is an additional relation Author(authorId, firstName, lastName) with 150 records, each 104 bytes (2 + 48 + 48 bytes).

Estimate the cost of performing a block nested-loop join between Employee and Department on DeptID, assuming 20 buffer blocks are available. Which relation should be the outer relation and why?